



Course: ENR 452L
Assignment: Lab 01
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1 Assigned questions

1. What value did you assign `_estack` in your linker file?

`_estack` was set to `0x24080000` for the M7 and `0x30020000` for the M4.

2. What value did you assign for the Flash Location and Length in your linker file?

The flash address is `0x08000000` on the M7 and `0x08100000` on the M4. Both have a length of 1024 KiB.

3. What value did you assign for the RAM Location and Length in your linker file?

The RAM address is `0x24000000` on the M7 (with a length of 512 KiB) and `0x30000000` on the M4 (with a length of 128 KiB).

4. What value did you assign to the `r7` register in the `reset_handler.s` file?

On the M7 core, `0xC0FFEE`. On the M4, `0xFACADE`.

5. What assembly instruction did you use for incrementing your counting register?

The ARM thumb spec defines the instruction `ADDS` to add an immediate value to a register while updating appropriate flags.

6. In which documents did you find this information? page?

ARM publishes a "quick reference card" for their thumb instruction set. The `ADDS` instruction is located on page 1 of this document.

7. In the linker files, you assign the RAM memory size and location that your program will use. If you have one program running on the Cortex-M4 MCU and one program running on the Cortex-M7 MCU, what change would you have to make to the linker files?

The programs would have to use different registers, different areas of flash, and different areas of RAM.

2 Running the code

As shown in Figures 1 and 2, the M7 core was loaded with a copy of the `reset_handler.s` script. This was executed, and stopped when `r0` was at `0x252ddb`, started again, and stopped when `r0` reached `0x5cd2e6a`. A similar process was completed for the M4 core, except using `r1` to count, and stopping at `0x252dc3b` and `0x441cd03`. This is shown in Figures 3 and 4.

```

(gdb) c
Continuing.
^C
Program received signal SIGTRAP, Trace/breakpoint trap.
0x08000010 in reset_handler ()
(gdb) i r
r0          0x252ddbd      38985149
r1          0x0           0
r2          0x0           0
r3          0x0           0
r4          0x8           8
r5          0x0           0
r6          0x0           0
r7          0xc0ffee      12648430
r8          0x0           0
r9          0x0           0
r10         0x0           0
r11         0x0           0
r12         0x0           0
sp          0x24080000     0x24080000
lr          0xffffffff     -1
pc          0x8000010      0x8000010 <reset_handler+8>
xpsr       0x1000000      16777216
msp        0x24080000     0x24080000
psp        0x0            0x0
control    0x0            0 '\000'
faultmask  0x0            0 '\000'
basepri    0x0            0 '\000'
primask    0x0            0 '\000'
fpscr      0x0            0

```

Figure 1: First snapshot of the registers while running a program on the M7 core.

```

(gdb) c
Continuing.
^C
Program received signal SIGTRAP, Trace/breakpoint trap.
0x08000010 in reset_handler ()
(gdb) i r
r0          0x5cd2e6a      97332842
r1          0x0           0
r2          0x0           0
r3          0x0           0
r4          0x8           8
r5          0x0           0
r6          0x0           0
r7          0xc0ffee      12648430
r8          0x0           0
r9          0x0           0
r10         0x0           0
r11         0x0           0
r12         0x0           0
sp          0x24080000     0x24080000
lr          0xffffffff     -1
pc          0x8000010      0x8000010 <reset_handler+8>
xpsr       0x1000000      16777216
msp        0x24080000     0x24080000
psp        0x0            0x0
control    0x0            0 '\000'
faultmask  0x0            0 '\000'
basepri    0x0            0 '\000'
primask    0x0            0 '\000'
fpscr      0x0            0

```

Figure 2: Second snapshot of the registers while running a program on the M7 core.

```

(gdb) load
Loading section .text, size 0x1c lma 0x8100000
Start address 0x08100000, load size 28
Transfer rate: 4 bytes/sec, 28 bytes/write.
(gdb) c
Continuing.
^C
Program received signal SIGTRAP, Trace/breakpoint trap.
i 0x08100010 in reset_handler ()
(gdb) i r
r0          0x30020000      805437440
r1          0x252dc3b       38984763
r2          0x0             0
r3          0x0             0
r4          0x0             0
r5          0x0             0
r6          0x0             0
r7          0xfacade       16435934
r8          0x0             0
r9          0x0             0
r10         0x0             0
r11         0x0             0
r12         0x0             0
sp          0x30020000      0x30020000
lr          0xffffffff      -1
pc          0x8100010       0x8100010 <reset_handler+8>
xpsr       0x1000000       16777216
msp        0x30020000      0x30020000
psp        0x0             0x0
control    0x0             0 '\000'
faultmask  0x0             0 '\000'
basepri    0x0             0 '\000'
primask    0x0             0 '\000'
fpscr      0x0             0
(gdb)

```

Figure 3: First snapshot of the registers while running a program on the M4 core.

```

(gdb) c
Continuing.
^C
Program received signal SIGTRAP, Trace/breakpoint trap.
0x08100010 in reset_handler ()
(gdb) i r
r0          0x30020000      805437440
r1          0x441cd03       71421187
r2          0x0             0
r3          0x0             0
r4          0x0             0
r5          0x0             0
r6          0x0             0
r7          0xfacade       16435934
r8          0x0             0
r9          0x0             0
r10         0x0             0
r11         0x0             0
r12         0x0             0
sp          0x30020000      0x30020000
lr          0xffffffff      -1
pc          0x8100010       0x8100010 <reset_handler+8>
xpsr       0x1000000       16777216
msp        0x30020000      0x30020000
psp        0x0             0x0
control    0x0             0 '\000'
faultmask  0x0             0 '\000'
basepri    0x0             0 '\000'
primask    0x0             0 '\000'
fpscr      0x0             0
(gdb)

```

gdb main.elf st-util

Figure 4: Second snapshot of the registers while running a program on the M4 core.